

POWER BI PROJECT — CUSTOMER CHURN ANALYSIS

Overall Project Meaning

Customer Churn Analysis means analyzing customers to understand:

- How many customers the company has
- How many customers are leaving the company
- Which type of customers are leaving
- Which services are associated with higher churn
- Which contract types have higher churn
- Which payment methods have higher churn
- Which customer characteristics are connected with churn



PAGE 1 — CUSTOMER DEMOGRAPHICS ANALYSIS

KPI Cards

1. Average Monthly Charges

→ 64.76

Meaning:

→ On average, a customer pays approximately 64.76 per month.

2. Total Customers

→ 7K

Meaning:

→ The company has approximately 7,000 customers.

3. Churned Customers

→ 1.869K

Meaning:

→ Approximately 1,869 customers have left the company.

4. Churn Rate

→ 0.27 → 27%

Meaning:

→ Approximately 27 out of every 100 customers are churning.

1. Customers Distribution by Gender

Chart type:

→ Donut Chart

What does it show?

→ This chart shows the distribution of customers by gender.

→ It compares Male customers vs Female customers.

Values:

→ Male → 4K → 50.4%

→ Female → 3K → 49.5%

Main observation:

→ Male and female customers are almost equally distributed.

→ Male customers are slightly higher.

Easy note:

“The chart shows the distribution of customers by gender. Male and female customers are almost equally distributed, with male customers being slightly higher.”

2. Average Monthly Charges by Tenure Group

Chart type:

→ Line Chart

What does it show?

→ This chart shows the average monthly charges for different customer tenure groups.

Tenure means:

→ How long the customer has been with the company.

X-axis:

→ Tenure Group

Examples:

→ 0–12 months

→ 13–24 months

→ 25–36 months

→ 37–48 months

→ 49–60 months

→ 61–72 months

Y-axis:

→ Monthly Charges

Trend:

→ 0–12 months → high charges

→ 13–24 months → charges decrease

→ 25–36 months → lower

→ 37–48 months → almost similar

→ 49–60 months → slightly increases

→ 61–72 months → increases again

Main observation:

→ Charges are high for new customers.

→ Charges decrease during the middle tenure groups.

→ Charges increase again for long-term customers.

Easy note:

“This chart shows how average monthly charges change according to customer tenure.”

3. Monthly Charges vs Tenure by Gender

Chart type:

→ Scatter Chart

What does it show?

→ This chart compares customer tenure and monthly charges based on gender.

X-axis:

→ Average Tenure

Y-axis:

→ Average Monthly Charges

Values visible:

→ Female → Average tenure ≈ 32.2 months → monthly charge ≈ 65

→ Male → Average tenure ≈ 32.4 months → monthly charge ≈ 64

Main observation:

→ Both genders have almost the same average tenure.

→ Female customers have slightly higher average monthly charges.

Easy note:

“The chart compares average tenure and monthly charges between male and female customers. Both genders have similar tenure, while females have slightly higher monthly charges.”

4. Customers by Tenure Group

Chart type:

→ Column Chart

What does it show?

→ This chart shows the number of customers in each tenure group.

Approximate values:

→ 0–12 months → 2.2K customers

→ 13–24 months → 1.0K

→ 25–36 months → 0.8K

→ 37–48 months → 0.8K

→ 49–60 months → 0.8K

→ 61–72 months → 1.4K

Main observation:

→ 0–12 months has the highest number of customers.

→ The middle tenure groups have fewer customers.

→ 61–72 months is the second-largest group.

Easy note:

“The chart shows the number of customers according to their tenure. The 0–12 month group has the highest number of customers.”

5. Total Customers by SeniorCitizen and Churn

Chart type:

→ 100% Stacked Column Chart

What does it show?

→ This chart compares senior citizens and non-senior citizens.

→ It also shows whether they churned or stayed.

SeniorCitizen = 0

→ No churn → approximately 75%

→ Churn → approximately 25%

SeniorCitizen = 1

→ No churn → approximately 60%

→ Churn → approximately 40%

Main observation:

→ Senior citizens have a higher proportion of churn.

→ Non-senior customers have a lower churn proportion.

Easy note:

“Senior citizens have a higher proportion of churn compared with non-senior customers.”

6. Total Customers by Tenure Group and Churn

Chart type:

→ 100% Stacked Column Chart

What does it show?

→ This chart shows the relationship between customer tenure and churn.

Trend:

→ 0–12 months → higher churn

→ 13–24 months → churn decreases

→ 25–36 months → decreases further

→ 37–48 months → lower churn

→ 49–60 months → lower churn

→ 61–72 months → lowest churn

Main observation:

→ New customers are more likely to leave.

→ As customers stay longer, their churn generally decreases.

Easy note:

“The chart shows that newer customers have higher churn, while long-term customers are more likely to stay.”



PAGE 2 — SERVICE SUBSCRIPTION ANALYSIS

KPI Cards

→ Average Monthly Charges → 64.76

→ Total Customers → 7K

→ Churned Customers → 1.869K

→ Churn Rate → 27%

1. Phone Service Distribution

Chart type:

→ Donut Chart

What does it show?

→ This chart shows customers who have phone service and those who don't.

Values:

→ Phone Service = Yes → 6K → 90.32%

→ Phone Service = No → 1K → 9.68%

Main observation:

→ Most customers have phone service.

→ Only a small percentage do not have phone service.

Easy note:

“Around 90% of customers have phone service, while around 10% do not.”

2. Churned Customers by Internet Service

Chart type:

→ Bar Chart

What does it show?

→ This chart shows the number of churned customers according to internet service type.

Values:

→ Fiber optic → 1.30K churned customers

→ DSL → 0.46K

→ No internet service → 0.11K

Main observation:

→ Fiber optic has the highest number of churned customers.

→ DSL is lower.

→ No internet service has the lowest number.

Easy note:

“Fiber optic customers have the highest number of churned customers.”

Important: The percentage displayed on this visual appears to be a formatting/calculation issue, so focus on the customer counts.

3. Churn Rate by Internet Service

Chart type:

→ 100% Stacked Bar Chart

What does it show?

→ This chart compares the churned and non-churned customers for each internet service type.

→ No → Customers who stayed

→ Yes → Customers who churned

Main observation:

→ Fiber optic has a higher churn proportion.

→ DSL has a moderate churn proportion.

→ No internet service has a lower churn proportion.

Easy note:

“Fiber optic customers have a relatively higher churn rate compared with other internet service groups.”

4. Customers by Internet Service Type

Chart type:

→ Column Chart

What does it show?

→ This chart shows the total number of customers using each internet service type.

Approximate values:

→ No internet service → 1.5K

→ DSL → 2.4K

→ Fiber optic → 3.2K

Main observation:

→ Fiber optic has the highest number of customers.

→ DSL is second.

→ No internet service is lowest.

Easy note:

“Fiber optic is the most commonly used internet service among customers.”

5. Internet Service Distribution

Chart type:

→ Horizontal Bar Chart

What does it show?

→ It shows the distribution of customers among different internet services.

→ Fiber optic → ~3.2K

→ DSL → ~2.4K

→ No internet service → ~1.5K

Main observation:

→ Fiber optic has the largest customer base.

Easy note:

“The chart shows that fiber optic has the highest number of customers.”

6. Services Count vs Churn

Chart type:

→ Scatter Chart

What does it show?

→ This chart studies the relationship between:

→ Number of services used

and

→ Churn rate

X-axis:

→ Services Count

Y-axis:

→ Churn Rate

General trend:

- Fewer services → generally higher churn
- More services → generally lower churn

Visible examples:

- Around 0 services → churn $\approx$ 45%
- Around 1 service → $\approx$ 22%
- Around 3 services → $\approx$ 35%
- Around 5 services → $\approx$ 24%
- Around 7 services → $\approx$ 8%

Main observation:

- Customers using more services generally have lower churn.

Easy note:

“Customers using more services generally have a lower churn rate, while customers using fewer services are more likely to churn.”



PAGE 3 — CONTRACT & BILLING INSIGHTS

KPI Cards

- Average Monthly Charges → 64.76
- Average Total Charges → 2.28K
- Churn Rate → 27%
- Churned Customers → 1.869K

1. Churn by Payment Method

Chart type:

- Donut Chart

What does it show?

→ This chart shows the number/proportion of churned customers according to their payment method.

Values visible:

→ Electronic check → 1.07K → 57.3%

→ Bank transfer → 0.31K → ~16%

→ Mailed check → 0.23K → ~12%

→ Credit card → remaining portion

Main observation:

→ Electronic check has the highest churn.

Easy note:

“Electronic check users have the highest number of churned customers.”

2. Average Tenure by Contract

Chart type:

→ Column Chart

What does it show?

→ This chart shows the average number of months customers stay with the company for each contract type.

Contract types:

→ Month-to-month

→ One year

→ Two year

Approximate values from the visual:

→ Month-to-month → ~19 months

→ One year → ~43 months

→ Two year → ~57 months

Main observation:

- Two-year contract customers have the highest average tenure.
- Month-to-month customers have the lowest average tenure.

Easy note:

“Customers with longer contracts generally stay with the company for a longer period.”

3. Average Total Charges by Contract

Chart type:

- Horizontal Bar Chart

What does it show?

- This chart compares the average total charges for each contract type.

Trend:

- Month-to-month → lowest average total charges
- One year → higher
- Two year → highest

Approximate values:

- Month-to-month → around 1.3K
- One year → around 2.8K
- Two year → around 3.7K

Main observation:

- Two-year contract customers have the highest average total charges.

Easy note:

“Customers with longer-term contracts generate higher average total charges.”

4. Churn Rate by Contract Type

Chart type:

→ Column Chart

What does it show?

→ This chart shows how churn changes according to contract type.

Approximate values:

→ Month-to-month → ~45–46%

→ One year → ~11%

→ Two year → ~3%

Main observation:

→ Month-to-month customers have very high churn.

→ Two-year customers have very low churn.

Easy note:

“Month-to-month contracts have the highest churn rate, while two-year contracts have the lowest churn rate.”

★ Important finding:

→ Long-term contract → Lower churn

5. Paperless Billing vs Churn

Chart type:

→ Stacked Column Chart

What does it show?

→ This chart compares customers who use paperless billing with customers who don't, and shows their churn status.

Categories:

→ Paperless Billing = No

→ Paperless Billing = Yes

Each bar contains:

- No → Customers who stayed
- Yes → Customers who churned

Main observation:

- Both groups contain customers who stayed and customers who churned.
- The visual allows us to compare churn between paperless-billing groups.

Easy note:

“This chart compares customer churn between paperless and non-paperless billing customers.”

6. Average Monthly Charges by Payment Method

Chart type:

- Column Chart

What does it show?

- This chart compares average monthly charges for different payment methods.

Payment methods:

- Mailed check
- Credit card
- Bank transfer
- Electronic check

Approximate trend:

- Mailed check → lower monthly charges
- Credit card → higher
- Bank transfer → higher

→ Electronic check → highest

Main observation:

→ Electronic check customers have the highest average monthly charges.

Easy note:

“The chart compares average monthly charges by payment method. Electronic check customers have the highest average monthly charges.”

PAGE 4 — CHURN PREDICTION & KEY DRIVERS

This is the most important page because it explains the major factors associated with customer churn.

Main KPI

→ Churn Rate → 0.27 → 27%

Meaning:

→ Out of every 100 customers, approximately 27 customers churn.

1. Revenue of Churn Status

Chart type:

→ Column Chart

What does it show?

→ This chart compares the total charges/revenue associated with customers who churned and customers who stayed.

Approximate values:

→ Churn = Yes → ~3M

→ Churn = No → ~13M

Main observation:

→ Customers who stayed generated much higher total charges.

Easy note:

“Non-churned customers contribute much more to total charges than churned customers.”

2. Churn Rate by Tenure Group

Chart type:

→ Column Chart

What does it show?

→ This chart shows how churn changes based on how long customers have been with the company.

Values:

→ 0–12 months → ~47%

→ 13–24 months → ~31%

→ 25–36 months → ~24%

→ 37–48 months → ~22%

→ 49–60 months → ~18%

→ 61–72 months → ~9%

Main observation:

→ New customers have the highest churn.

→ Churn decreases as tenure increases.

Easy note:

“Customers with shorter tenure have a higher churn rate, while long-term customers have a much lower churn rate.”

★ Key driver:

→ Short tenure → High churn

3. Churn Rate by SeniorCitizen

Chart type:

→ Horizontal Bar Chart

What does it show?

→ This chart compares churn between senior and non-senior customers.

Values:

→ Senior Citizen = 1 → ~50%

→ Senior Citizen = 0 → ~15%

Main observation:

→ Senior citizens have a much higher churn rate.

Easy note:

“Senior citizens have a significantly higher churn rate than non-senior customers.”

★ Key driver:

→ Senior citizen → Higher churn

4. Churn Rate by Dependents

Chart type:

→ Horizontal Bar Chart

What does it show?

→ This chart compares customers with dependents and without dependents.

Values:

→ Dependents = Yes → ~19%

→ Dependents = No → ~34%

Main observation:

→ Customers without dependents have higher churn.

Easy note:

“Customers without dependents are more likely to churn than customers with dependents.”

5. Churn Rate by Contract Type

Chart type:

→ Horizontal Bar Chart

Values:

→ Month-to-month → ~47%

→ One year → ~11%

→ Two year → ~3%

Main observation:

→ Month-to-month customers are much more likely to leave.

→ Two-year contract customers are much more likely to stay.

Easy note:

“Month-to-month contracts have the highest churn rate, while two-year contracts have the lowest churn rate.”

★ Major key driver:

→ Month-to-month contract → High churn

6. Churn Rate by Partner

Chart type:

→ Horizontal Bar Chart

What does it show?

→ It compares customers with a partner and without a partner.

Values:

→ Partner = Yes → ~19%

→ Partner = No → ~32%

Main observation:

→ Customers without partners have higher churn.

Easy note:

“Customers without partners have a higher churn rate than customers with partners.”

7. Churn Rate by Internet Service

Chart type:

→ Horizontal Bar Chart

Values:

→ Fiber optic → ~42%

→ DSL → ~19%

→ No internet service → ~11%

Main observation:

→ Fiber optic customers have the highest churn rate.

Easy note:

“Fiber optic customers have the highest churn rate among the internet service groups.”

★ Key driver:

→ Fiber optic → Higher churn

8. Churn Rate by Payment Method

Chart type:

→ Horizontal Bar Chart

Values:

→ Electronic check → ~45%

→ Mailed check → ~20%

→ Bank transfer → ~18%

→ Credit card → ~16%

Main observation:

→ Electronic check customers have the highest churn rate.

→ Credit card customers have the lowest among these payment methods.

Easy note:

“Electronic check users have the highest churn rate compared with other payment methods.”

Key driver:

→ Electronic check → Higher churn

9. Contract × Internet Service Table

This is an important table because it combines two factors:

→ Contract type

AND

→ Internet service

Month-to-month

→ DSL → 32%

→ Fiber optic → 55%

→ No internet service → 19%

→ Total → 43%

One year

→ DSL → 9%

→ Fiber optic → 19%

→ No internet service → 2%

→ Total → 11%

Two year

→ DSL → 2%

→ Fiber optic → 7%

→ No internet service → 1%

→ Total → 2%

Overall

→ DSL → 19%

→ Fiber optic → 42%

→ No internet service → 7%

→ Overall → 27%

Most important finding:

→ Month-to-month + Fiber optic → 55% churn

This is the highest churn combination shown in the table.

★ MOST IMPORTANT FINDINGS FROM THE WHOLE PROJECT

“What did you find from your Power BI dashboard?”

You can say:

→ Total customers → 7K

→ Churned customers → 1.869K

→ Overall churn rate → 27%

→ Male and female customers are almost equally distributed.

→ Most customers have phone service → approximately 90%.

- Fiber optic has the highest number of internet-service customers.
- Fiber optic also has a high churn rate → approximately 42%.
- New customers have a higher churn rate.
- 0–12 month customers have approximately 47% churn.
- 61–72 month customers have approximately 9% churn.
- Month-to-month contracts have very high churn → approximately 47%.
- Two-year contracts have very low churn → approximately 3%.
- Senior citizens have higher churn.
- Customers without dependents have higher churn.
- Customers without partners have higher churn.
- Electronic check users have higher churn → approximately 45%.
- Month-to-month + Fiber optic customers have the highest combined churn → approximately 55%.

BEST FINAL PRESENTATION ANSWER

If your teacher asks you to explain your entire dashboard, you can say:

“My Power BI project is based on Customer Churn Analysis. The dashboard contains four pages: Customer Demographics Analysis, Service Subscription Analysis, Contract and Billing Insights, and Churn Prediction and Key Drivers.

The company has around 7,000 customers, of which approximately 1,869 customers have churned. The overall churn rate is 27%.

From the demographic analysis, male and female customers are almost equally distributed. The analysis also shows that new customers have a higher churn rate compared with long-term customers.

From the service analysis, most customers use phone service, and fiber optic is the most common internet service. Fiber optic customers also show a relatively high churn rate.

From the contract and billing analysis, month-to-month customers have a much higher churn rate than one-year and two-year contract customers. Electronic check users also have a higher churn rate.

Finally, the churn prediction page identifies the major factors associated with churn. Short tenure, month-to-month contracts, fiber optic internet service, electronic check payment, senior-citizen status, and not having dependents or a partner are associated with higher churn.

Therefore, the company can focus on new customers and month-to-month customers, improve customer retention, and encourage customers to move to longer-term contracts.”

Customer Churn Analysis — Simple Meanings

1. Customer

→ A person who uses or pays for a company's service.

Example: A person using Jio internet is a Jio customer.

2. Churn

→ When a customer leaves or stops using the company's service.

Example: A customer cancels their internet connection → Churn.

3. Churned Customer

→ A customer who has already left the company.

4. Churn Count

→ The number of customers who left the company.

Example: 200 customers left → Churn Count = 200.

5. Churn Rate

→ The percentage of customers who left the company.

Example: 27 out of 100 customers left → Churn Rate = 27%.

6. Tenure

→ The amount of time a customer has been using the company's service.

Example: Customer used the service for 2 years → Tenure = 24 months.

7. Tenure Group

→ A group of customers based on how long they have used the service.

Example: 0–12 months, 13–24 months, 25–36 months.

8. Service

→ Something that a company provides to customers.

Example: Internet, phone, online security.

9. Service Type

→ The different types of services offered by the company.

Example: DSL, Fiber optic, No internet service.

10. Internet Service

→ The type of internet connection a customer uses.

Example: DSL or Fiber optic.

11. DSL

→ A type of internet connection that uses telephone-line technology.

Copper wire internet came through it example: tv wire

12. Fiber Optic

→ A type of internet connection that uses fiber cables to provide internet. The internet passes through light very speed the wire is very thin

13. Phone Service

→ A service that allows customers to make and receive phone calls.

14. Services Count

→ The number of services used by one customer.

Example: Internet + Phone + Security = Services Count of 3.

15. Senior Citizen

→ An older adult/customer.

In this dataset: 1 = Senior Citizen, 0 = Not a Senior Citizen.

16. Dependent

→ A person who depends on another person for financial or personal support.

Example: A child who depends on their parent.

17. Partner

→ A customer's spouse or life partner.

Example: Husband, wife, or life partner.

18. Contract

→ An agreement between the customer and the company about the service.

19. Contract Type

→ The length/type of the customer's agreement with the company.

20. Month-to-Month Contract

→ The customer uses the service one month at a time without a long-term contract.

21. One-Year Contract

→ The customer has an agreement with the company for one year.

22. Two-Year Contract

→ The customer has an agreement with the company for two years.

23. Payment Method

→ The way a customer pays the company.

Example: Credit card, bank transfer, electronic check.

24. Electronic Check

→ A digital payment made from a bank account.

25. Bank Transfer

→ Money is transferred from the customer's bank account to the company.

26. Mailed Check

→ A customer sends a physical check by mail to the company.

27. Credit Card

→ A card used to pay for the service using credit.

28. Paperless Billing

→ Receiving the bill digitally instead of receiving a paper bill.

Example: Bill received through email/app.

29. Monthly Charges

→ The amount a customer pays every month for the service.

30. Total Charges

→ The total amount charged to a customer during their service period.

31. Average Monthly Charges

→ The average amount customers pay per month.

32. Average Total Charges

→ The average total amount charged to customers during their service period.

33. Churn Status

→ Shows whether the customer left or stayed.

→ Yes = Left

→ No = Stayed

34. Revenue

→ The money earned by the company from customers.

35. Subscription

→ When a customer signs up to use a service, usually by paying regularly.

36. Demographics

→ Basic information about customers.

Example: Gender, senior-citizen status, partner, dependents.

37. Customer Demographics

→ Information about who the customers are.

38. Churn Prediction

→ Finding customers who may be more likely to leave in the future.

39. Key Drivers

→ The main factors connected with customer churn.

40. Churn Driver

→ Something that is associated with a higher chance of customers leaving.

Example: Month-to-month contract → higher churn.

Tenure → How long they stayed

Service → What they use

Contract → What agreement they have

Payment Method → How they pay

Partner → Who they live/share life with

Dependent → Who depends on them

Churn → Whether they leave

Churn Count → How many left

Churn Rate → What percentage left

Key Driver → What is associated with customers leaving

You can copy this entire section directly into your notes.